# Security Analysis of Microscaling Formats Under Fault Injection on a RISC-V Edge Platform

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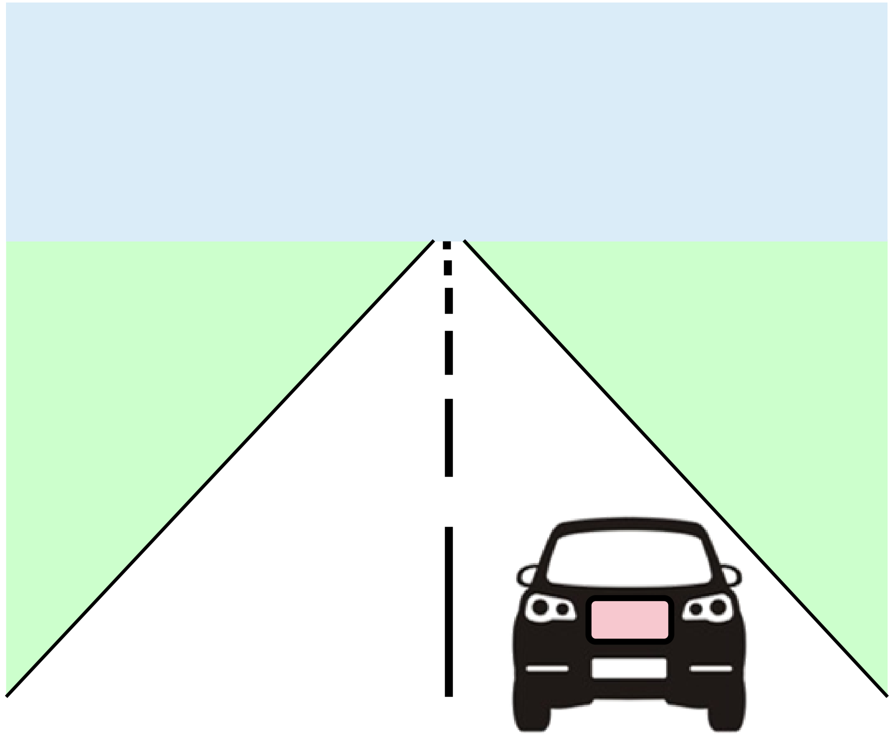
# Edge inference is rising

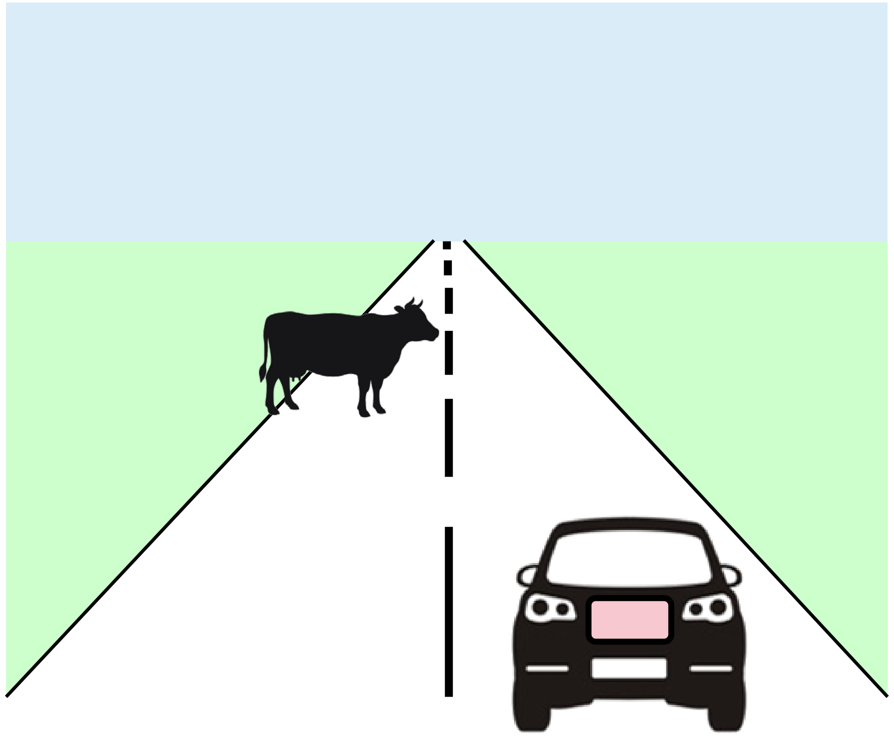


- Cameras
- Lidar
- Telemetry
- Navigation

**Self-driving AI**

- Throttle
- Steering
- Braking

# Edge inference implementation factors are challenging



- Cameras
- Lidar
- Telemetry
- Navigation

**Self-driving AI**

- Throttle
- Steering
- Braking

**Real-time**

**Resource-constrained**

**High-reliability**

# Microscaling format to support resource constraints

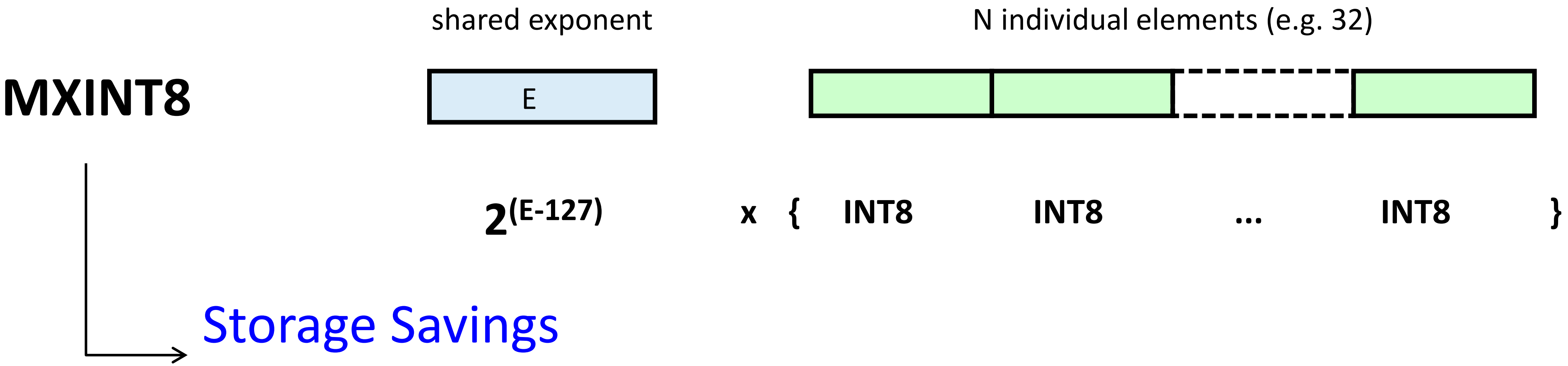
Real-time

**Resource-constrained: Clock Cycles x [Logic Gates, Storage]**

High-reliability

# Microscaling format: OCP MX Standard

Real-time
Resource-constrained: Clock Cycles x [Logic Gates, Storage]
High-reliability



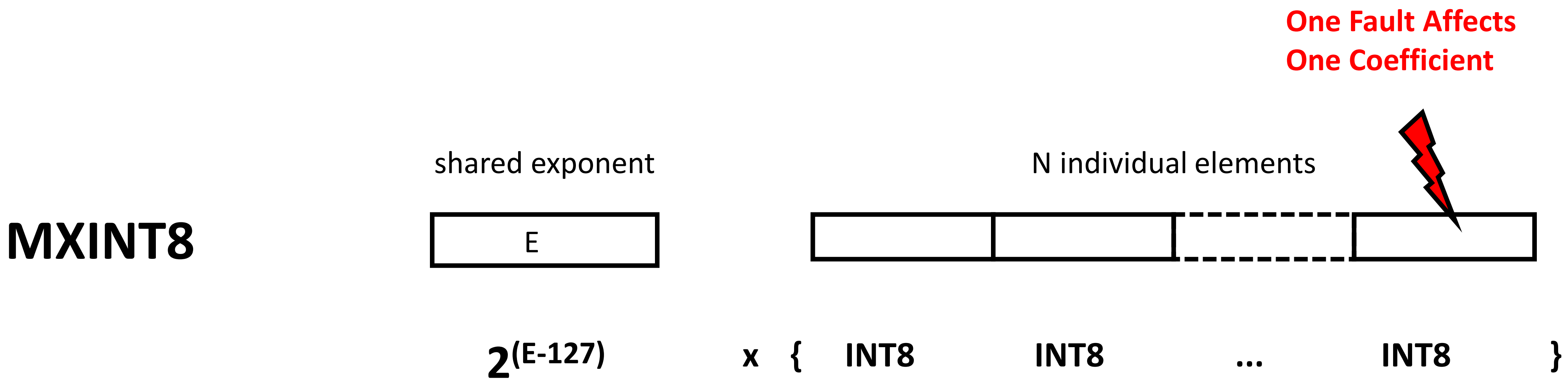

Storage Savings
Clock Cycle/Gate Savings with SIMD

# Microscaling format impacts reliability

Real-time

Resource-constrained: Clock Cycles x [Logic Gates, Storage]

**High-reliability**



# Microscaling format impacts reliability

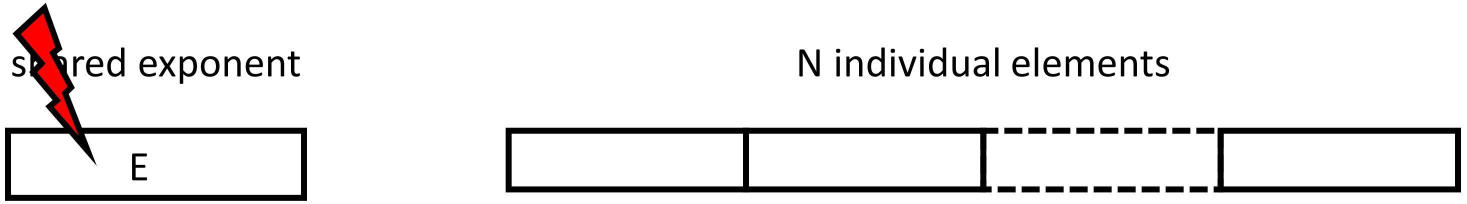
Real-time

Resource-constrained: Clock Cycles x [Logic Gates, Storage]

**High-reliability**

**One Fault Affects**
**N Coefficients**

**MXINT8**




$$2^{(E-127)} \quad \times \quad \{ \quad \text{INT8} \quad \text{INT8} \quad \ldots \quad \text{INT8} \quad \}$$

# Objectives: Evaluate reliability impact on a prototype

- Prototype Coprocessor for MX Processing
- Fault-injection Campaign on various workloads
- Post-fault Analysis at multiple levels

**Hypothesis**

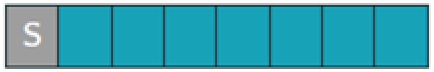
1. The shared-exponent of MX is sensitive to fault injection
2. Carefully designed MX can offer containment

# Prototype Experimental Platform

# We tested 5 MX formats

**Shared SE + N Fields of**

INT : linear




**OCP MXINT8**

FP : exponential


**OCP MXFP8/E5M2**
**OCP MXFP8/E4M3**

LOG : log-domain


**Proposed in this work**

**Open Compute Project**

# We tested 5 MX formats

Shared SE + N Fields of

| | INT : linear | FP : exponential | LOG : log-domain |
|---|---|---|---|
| | S \| \| \| \| \| \| \| | S \| E \| E \| E \| E \| M \| M \| M | S \| FI \| FI \| FI \| FI \| FE \| FE \| FE |
| ↔ Range | $\pm 127$, uniform | E4M3 $\pm 448$<br>E5M2 $\pm 57344$ | up to $\sim 2^{16}$,<br>log-spaced |
| Advantage | full accuracy,<br>robust | widest dynamic range | cheap mul |
| Variants | MXINT8 | MXFP8-E4M3,<br>MXFP8-E5M2 | LOG8-SUM,<br>LOG8-MAX |

# We tested 5 MX formats

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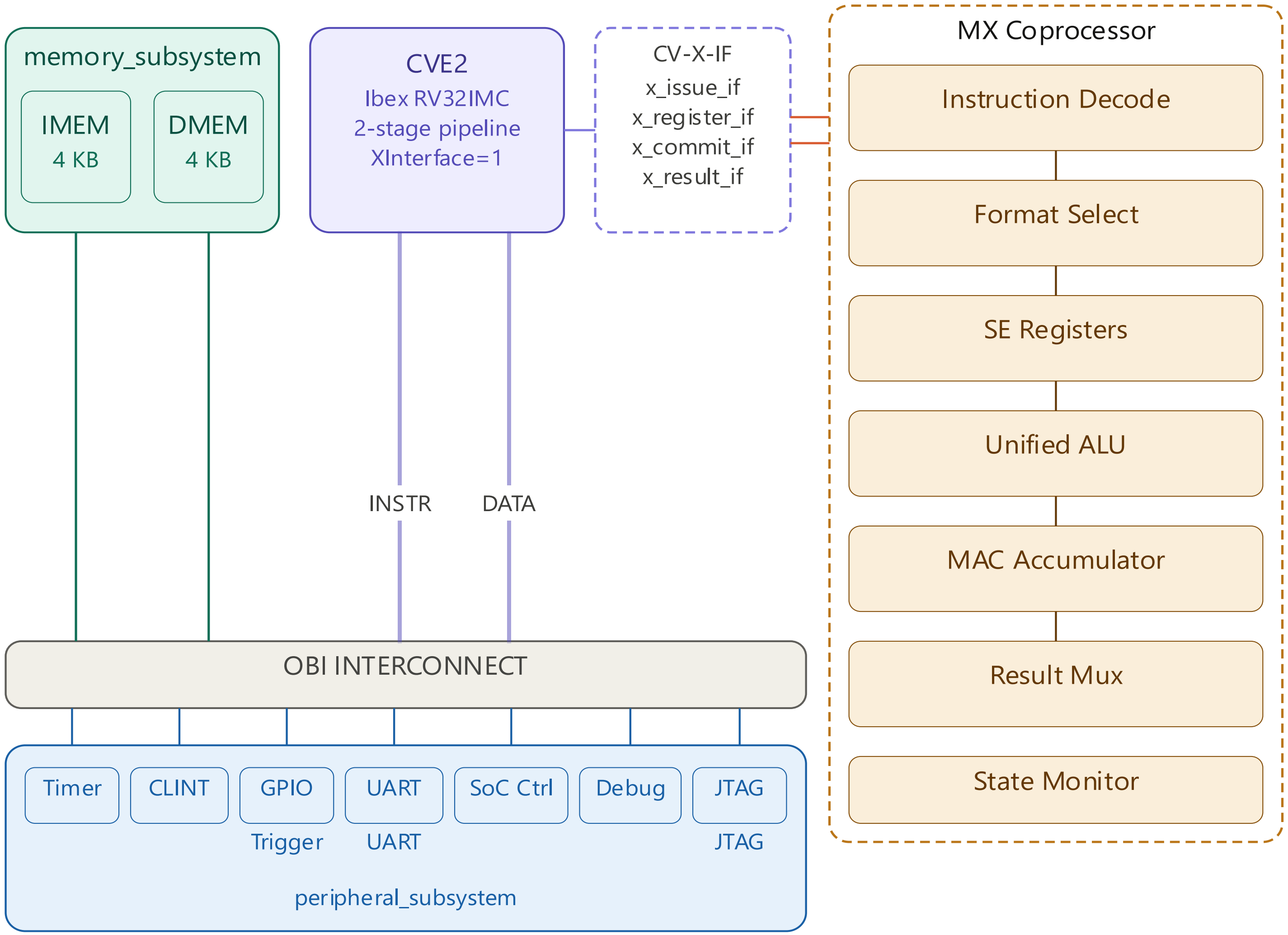
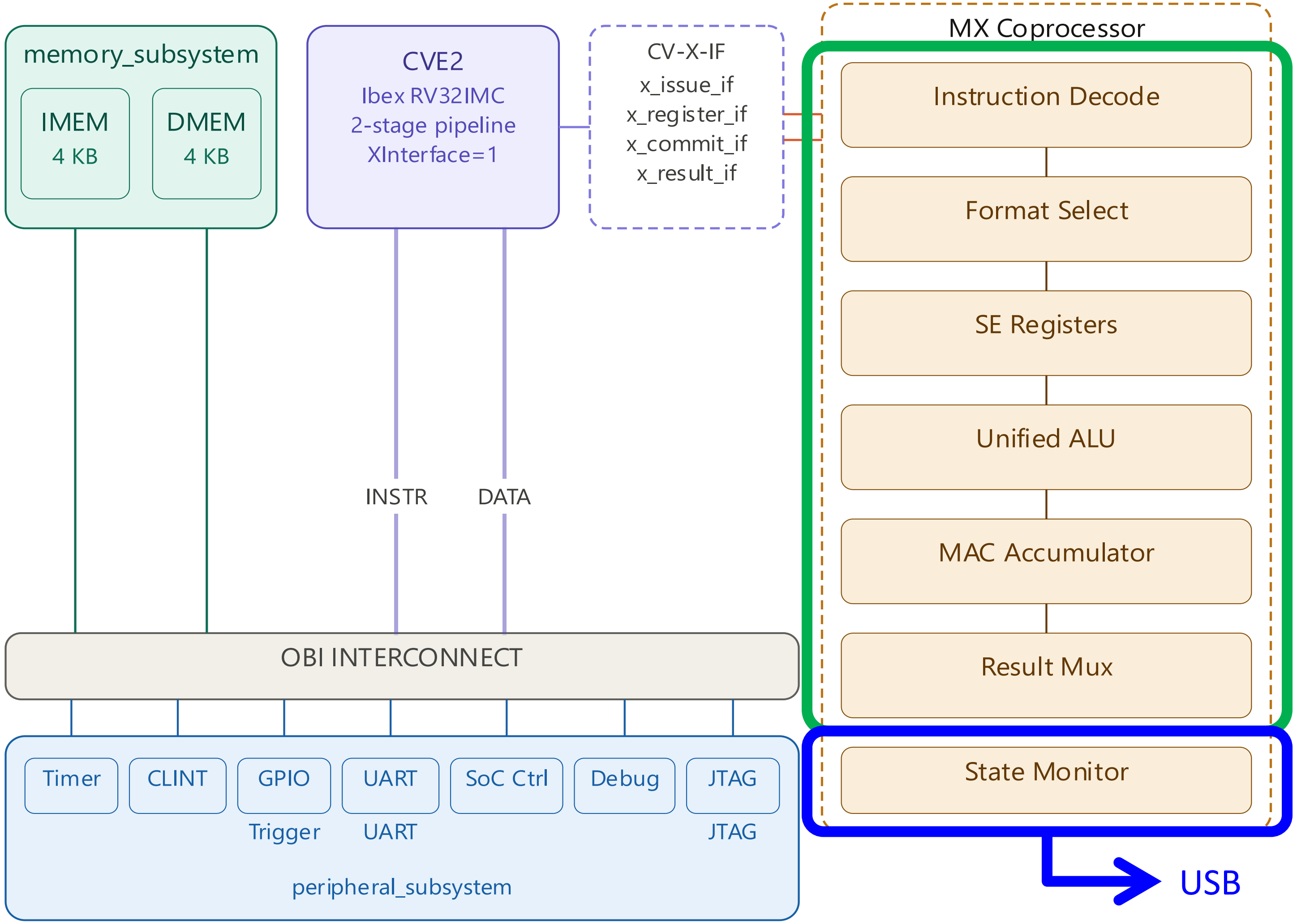
| Variants | MXINT8 | MXFP8-E4M3,<br>MXFP8-E5M2 | LOG8-SUM,<br>LOG8-MAX |
| **Sensitivity** | **128** | **256x**<br>**65,536x** | **256x**<br>**or *absorbed*** |

# Platform overview




- **RISC-V SoC with Custom-Instruction Acceleration**

# MX coprocessor




- **16 Inst across 5 MX formats**

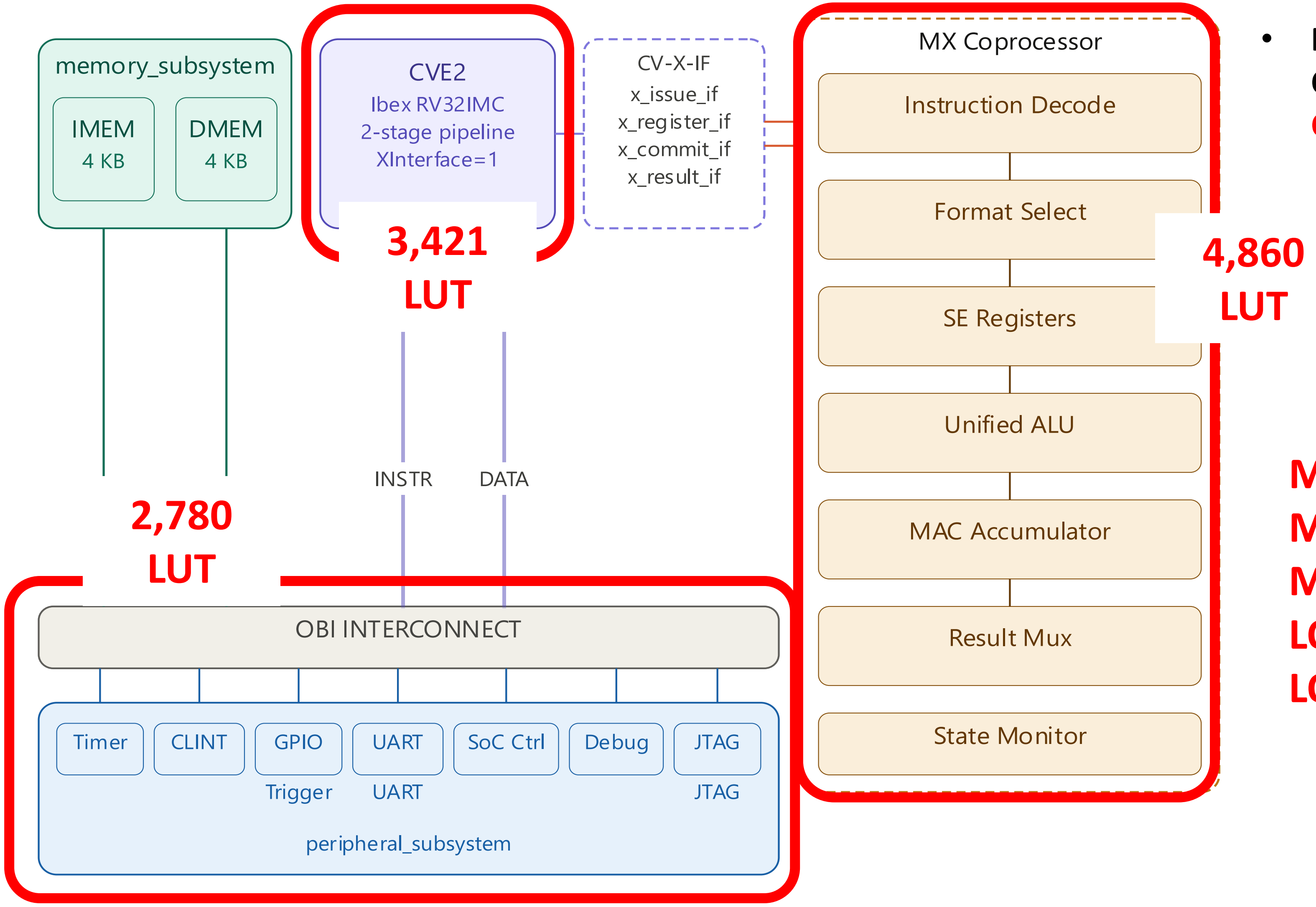
  SIMD Operations (on 4 elements)
  MAC Operations
  Conversion/Decode Operations

- **State Monitor 129 bits**

  Distinguish CPU from CP faults

  32-bit operand rs1, rs2
  32-bit MAC
  8-bit SE_a, SE_b
  4-bit opcode
  3-bit format selector
  1-bit flag
  5-bit destination reg
  3-bit instruction ID

# Platform area cost



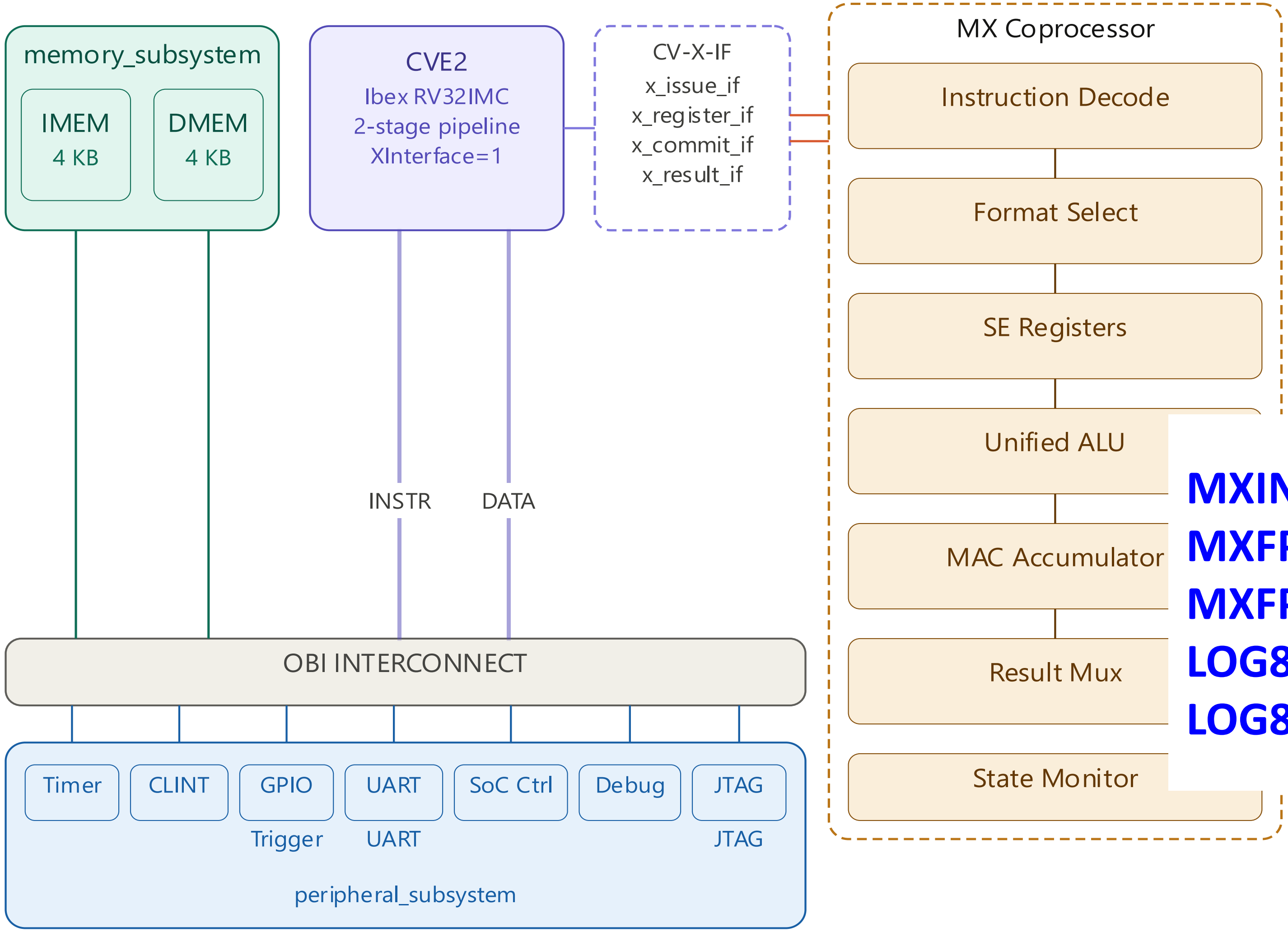

- **RISC-V SoC with Custom-Instruction Acceleration**
  **CW305 FPGA Implementation**

**Unifed area cost**

**Per-format area cost**

| | |
|---|---|
| MXINT8 | 1,019 |
| MXFP8-E4M3 | 2,105 |
| MXFP8-E5M2 | 2,178 |
| LOG8-SUM | 1,578 |
| LOG8-MAX | 905 |

# MX coprocessor performance



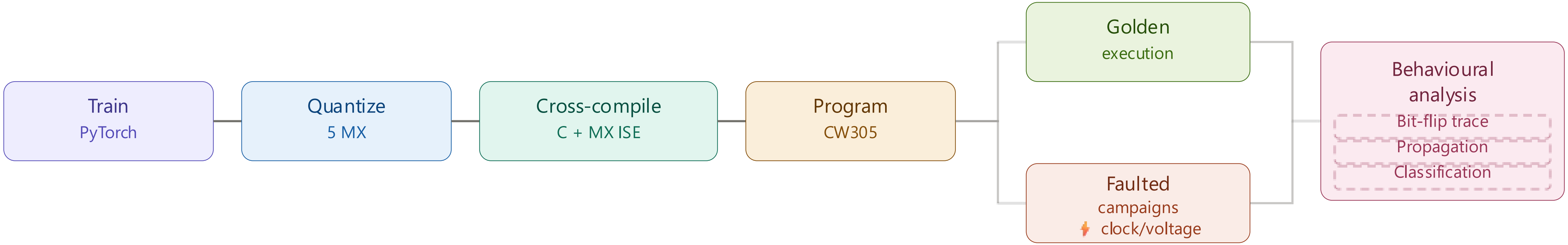

- **RISC-V SoC with Custom-Instruction Acceleration**

## DOT4 Performance

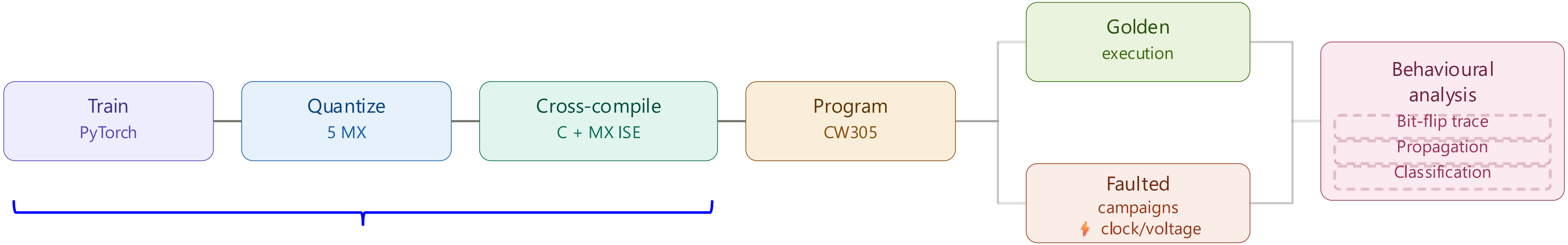
| | SW RV32IMC | HW |
|---|---|---|
| MXINT8 | 50-150 | 2 |
| MXFP8-E4M3 | 220 | 2 |
| MXFP8-E5M2 | 220 | 2 |
| LOG8-SUM | 82 | 2 |
| LOG8-MAX | 36 | 2 |

# Fault-Injection Campaign

# Fault injection pipeline



# 2 Workloads evaluated



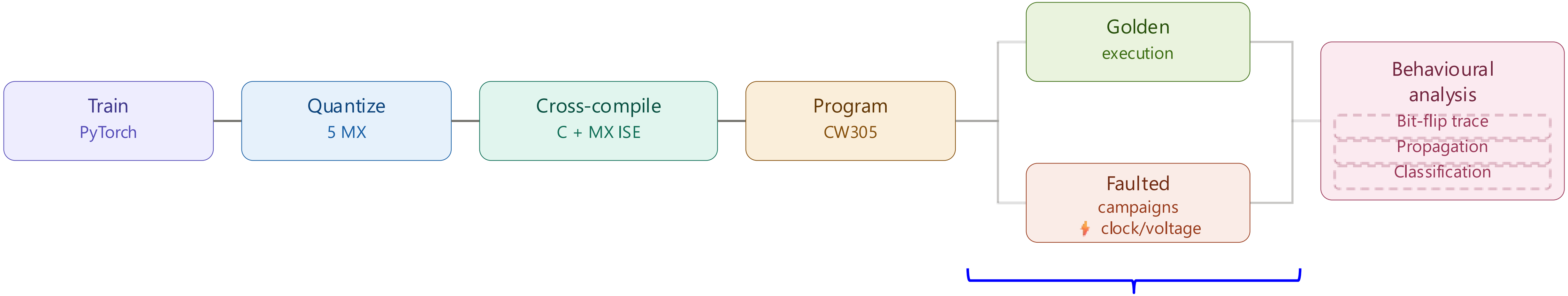

1

**QNN Workload**
**2-layer transformer decoder, 545 weights**
- **multilayer fault propagation through softmax**
- **format dependence of attention routing**

2

**BitNet b1.58 FC1 Workload, 2048 weights**
**1-layer 64-in, 32 neuron, softmax with ternary weights**
- **supports fault isolation and bit-position sensitivity analysis**

# Fault injection vectors




**1** **Voltage Underscaling**

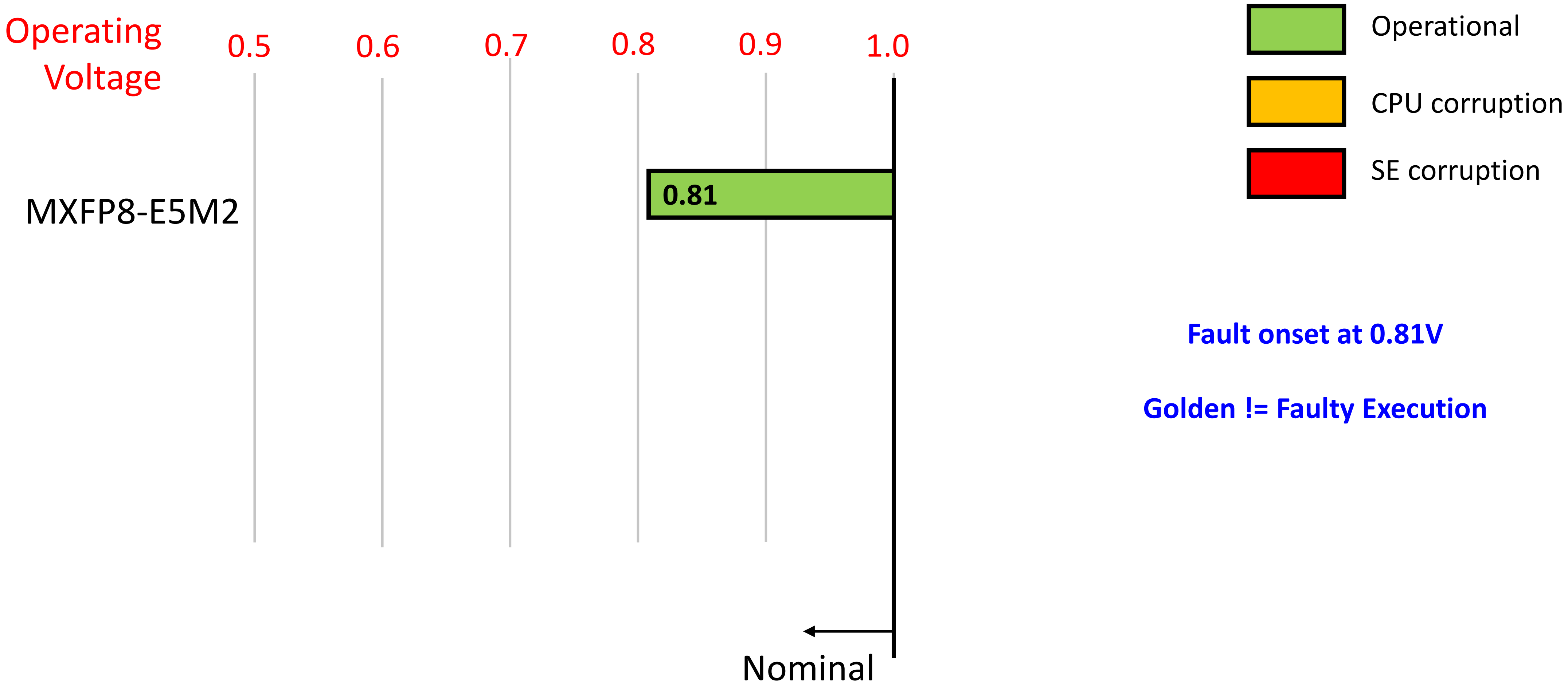
- **Reduce VCCint in 10mV steps**
- **Affects every clock cycle and overall architecture**

**2** **Clock Glitching**

- **Affect coprocessor execution with glitch sweep**

⇨ *No impact observed*

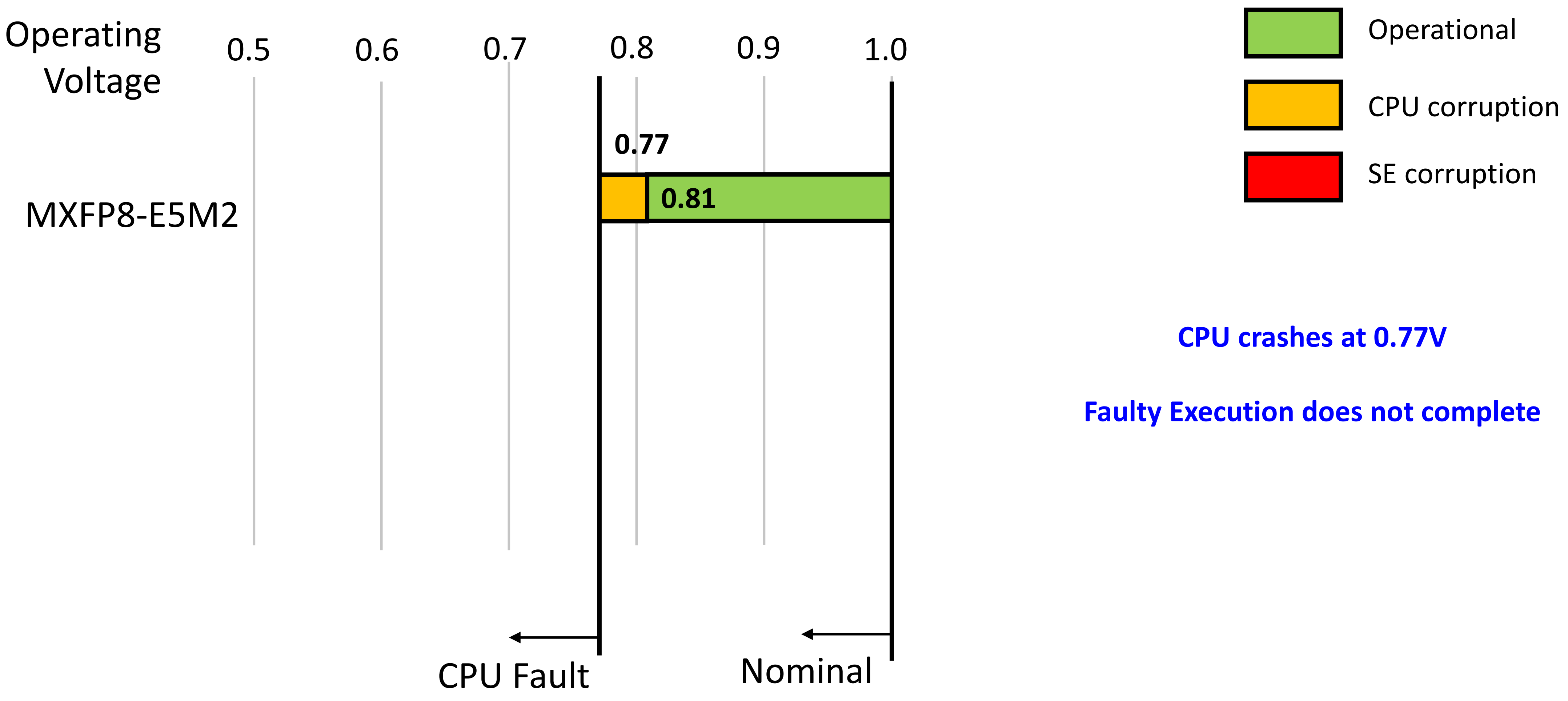
# Undervolting fault statistics for QNN workload




**Fault onset at 0.81V**

**Golden != Faulty Execution**

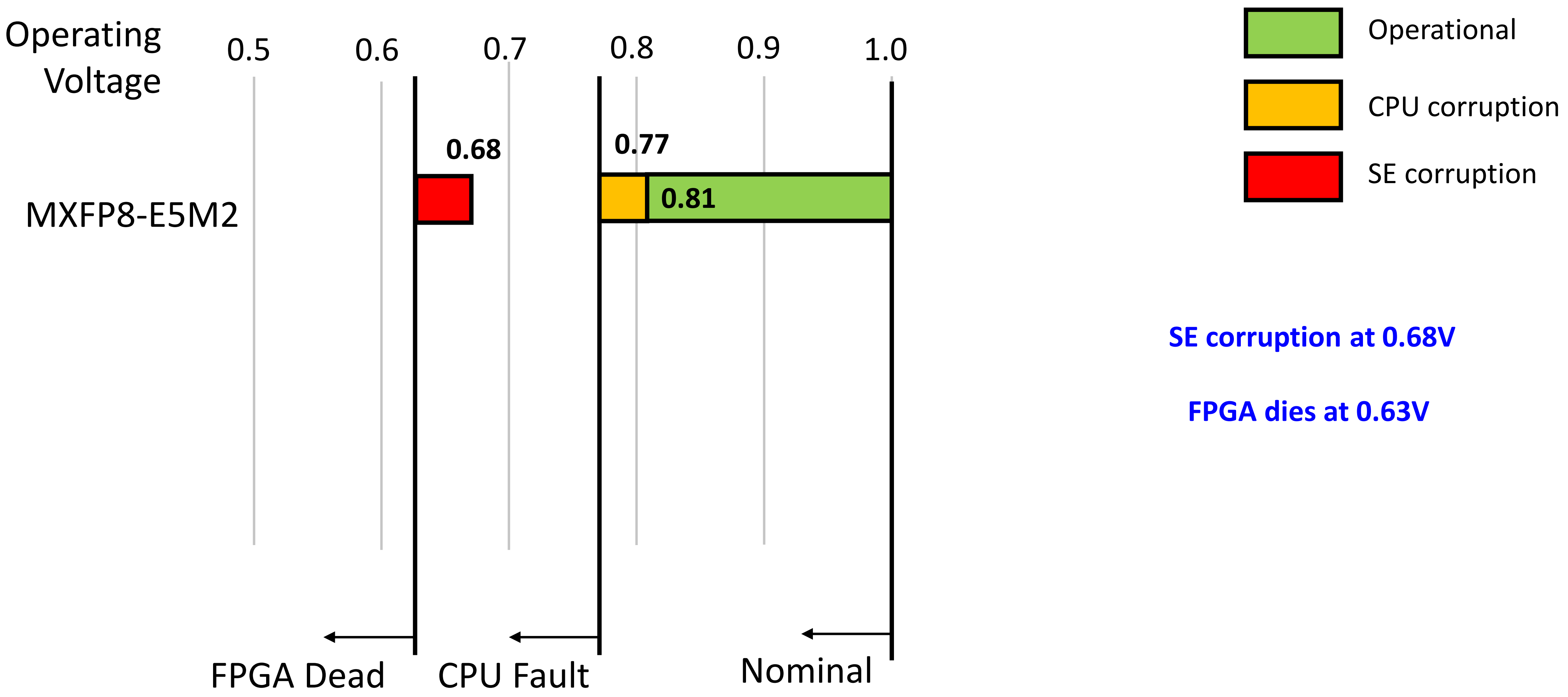
# Undervolting fault statistics for QNN workload




**CPU crashes at 0.77V**

**Faulty Execution does not complete**

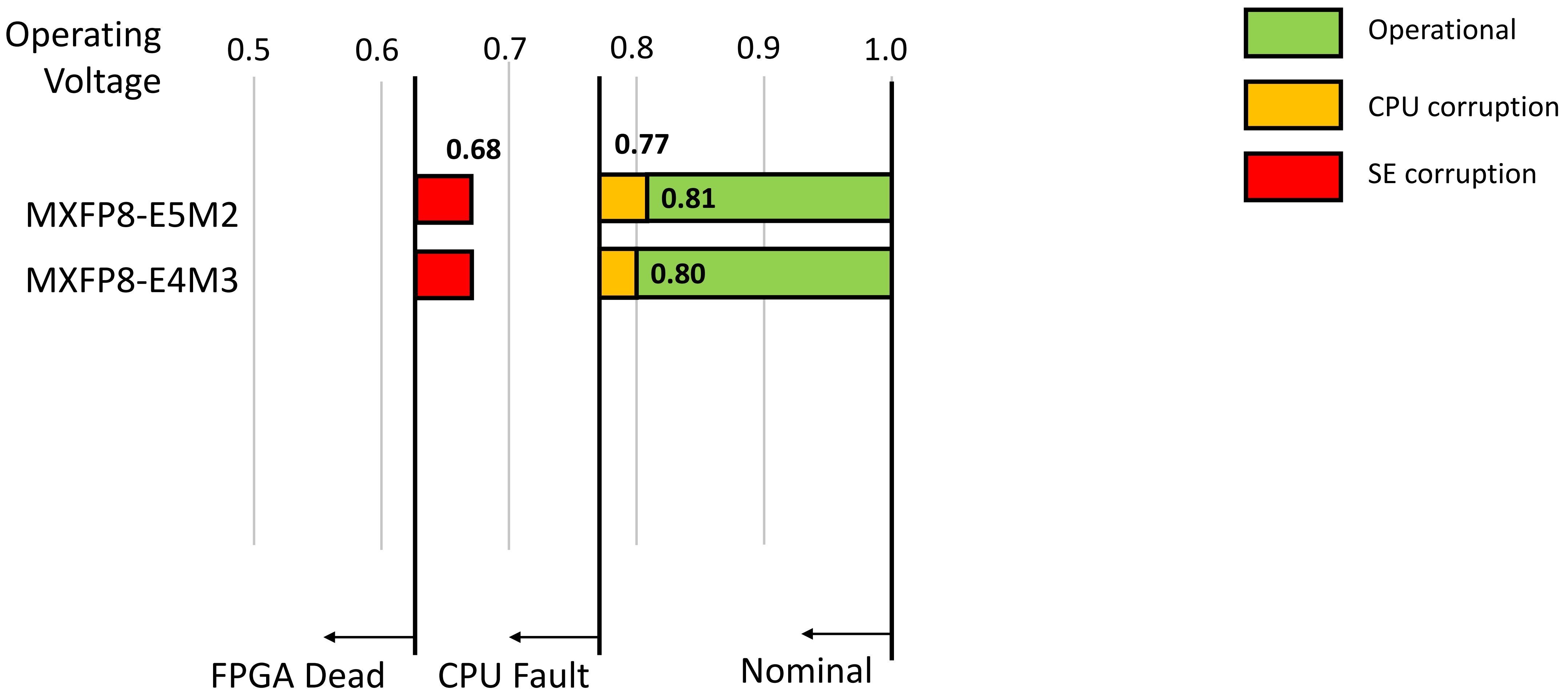
# Undervolting fault statistics for QNN workload




**SE corruption at 0.68V**

**FPGA dies at 0.63V**

# Undervolting fault statistics for QNN workload



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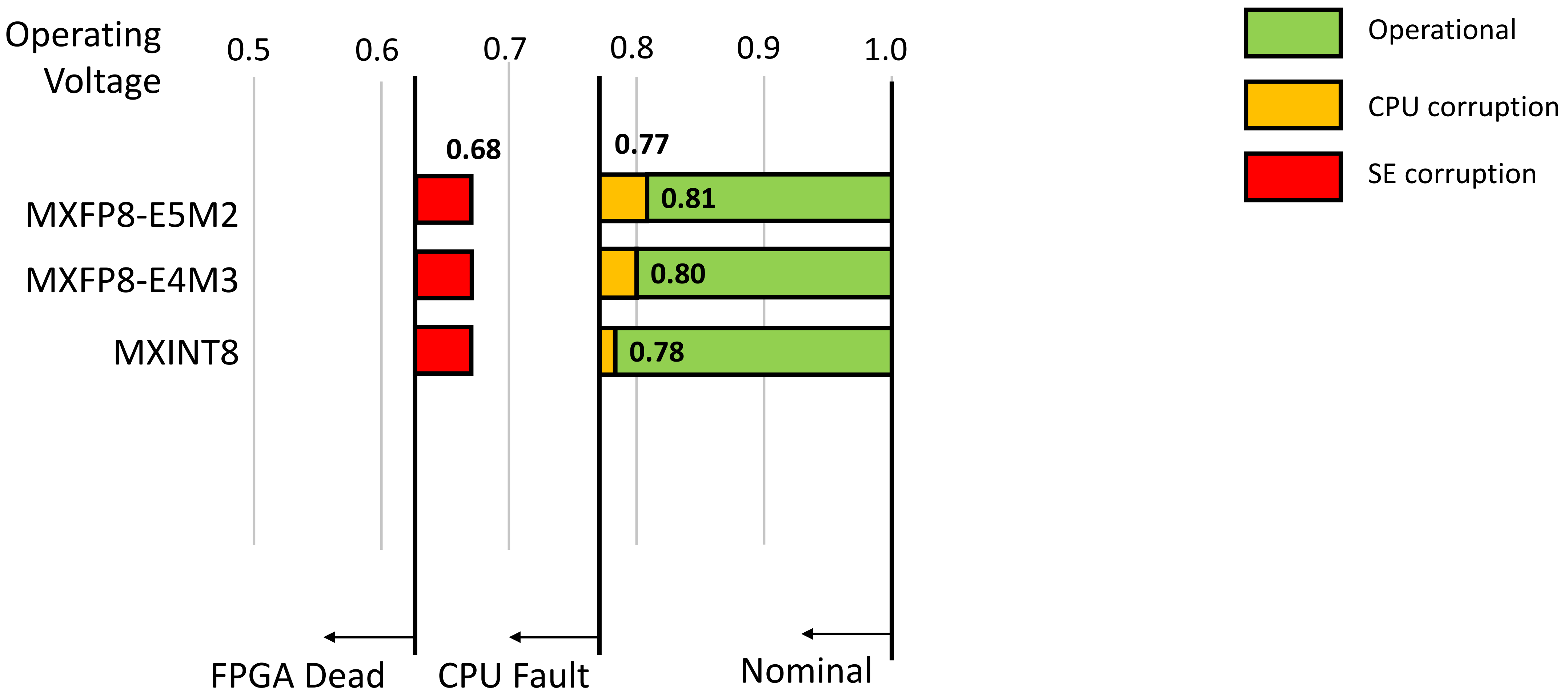

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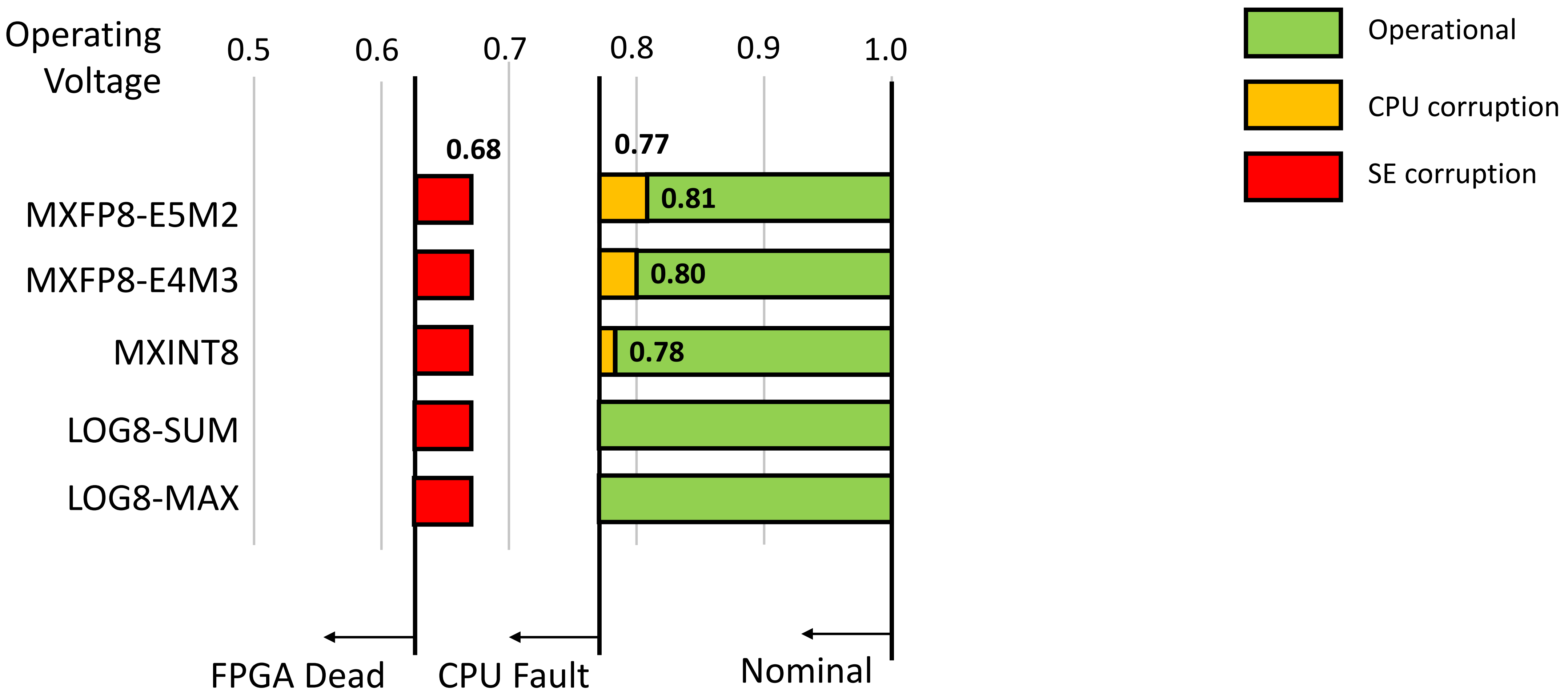
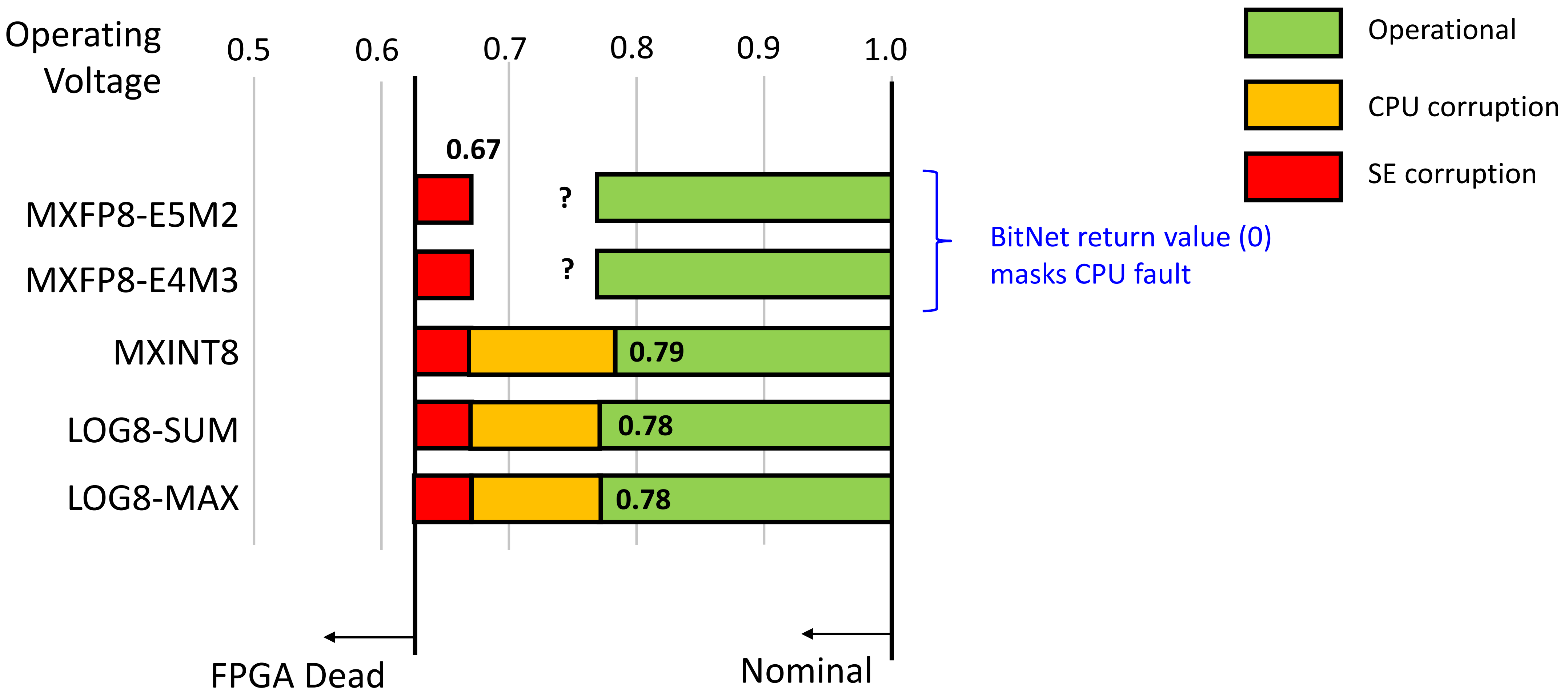


# Undervolting fault statistics for BitNet workload



# Hypothesis Testing

# H1: The shared-exponent of MX is sensitive to fault injection

- Confirmed in BitNet
  - At 0.67V, SE becomes corrupted (from 127 to 255) while CPU still accepts the result

# H2: Carefully designed MX can offer containment

- Confirmed by analyzing the number of faults in QNN during Voltage sweep

| Format | Normal | Faults | Crashes | Dead |
|---|---|---|---|---|
| MXFP8-E5M2 | 57 | 27 | 27 | 3 |
| MXFP8-E4M3 | 60 | 24 | 27 | 3 |
| MXINT8 | 66 | 18 | 27 | 3 |
| LOG8-SUM | 69 | **15** | 27 | 3 |
| LOG8-MAX | 69 | **15** | 27 | 3 |

LOG8-SUM Shortest critical path for dot product
LOG8-MAX Only faults in the max term survive

# Conclusions

# Conclusions

- Shared exponent corruption is relevant regardless of MX format
- Our experiments are open-source
  - https://github.com/Secure-Embedded-Systems/FI-Croc-XIF-UMX-CW305
- Future Work
  - Evaluate hypothesis on larger networks (QNN 548 weights; BitNet 2048 weights)
  - Optimize applications for LOG8-SUM and LOG8-MAX (avoid low accuracy on dense networks)

- Read our paper:
  - Proposed countermeasures
  - Design space summary (hardware cost - performance – latency – fault resilience)

# Thank you